EXPERIMENT 9

Build a text generation model using Long Short-Term Memory (LSTM) networks. Train the model on a text corpus to generate coherent sequences of text and evaluate the generated text for fluency and coherence.

```
import tensorflow as tf
import numpy as np
import urllib.request

# Download Shakespeare text
url = "https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt"
text = urllib.request.urlopen(url).read().decode("utf-8")

text = text[:100000]

# Create character vocabulary
vocab = sorted(set(text))
char_to_id = {c: i for i, c in enumerate(vocab)}
id_to_char = np.array(vocab)

encoded = np.array([char_to_id[c] for c in text])

# Create training sequences
seq_length = 40

dataset = tf.keras.utils.timeseries_dataset_from_array(
    encoded[:-1],
    targets=encoded[seq_length:],
    sequence_length=seq_length,
    batch_size=64
)

# Create LSTM model
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(len(vocab), 64),
    tf.keras.layers.LSTM(128, return_sequences=True),
    tf.keras.layers.LSTM(128),
    tf.keras.layers.Dense(len(vocab))
])

model.compile(
    optimizer="adam",
    loss=tf.keras.losses.SparseCategoricalCrossentropy(
        from_logits=True
    ),
    metrics=["accuracy"]
)

# Train model
print("Training LSTM model...")
model.fit(dataset, epochs=5)

# Generate text
def generate_text(start_text, length=300):
    result = start_text

    for _ in range(length):
        input_text = result[-seq_length:]
        input_ids = np.array(
            [[char_to_id[c] for c in input_text]]
        )

        predictions = model(input_ids, training=False)
        predictions = predictions[0] / 1.0

        predicted_id = tf.random.categorical(
            predictions[None, :], num_samples=1
        )[0, 0].numpy()

        result += id_to_char[predicted_id]

    return result

print("\nGenerated Text:\n")
print(generate_text("ROMEO: ", 300))
```

```
Training LSTM model...
Epoch 1/5
```

```
1562/1562 ━━━━━━━━━━━━━━━━━━━━ 295s 184ms/step - accuracy: 0.3055 - loss: 2.5103
Epoch 2/5
1562/1562 ━━━━━━━━━━━━━━━━━━━━ 283s 181ms/step - accuracy: 0.4080 - loss: 2.0560
Epoch 3/5
1562/1562 ━━━━━━━━━━━━━━━━━━━━ 281s 180ms/step - accuracy: 0.4510 - loss: 1.8843
Epoch 4/5
1562/1562 ━━━━━━━━━━━━━━━━━━━━ 286s 183ms/step - accuracy: 0.4804 - loss: 1.7688
Epoch 5/5
1562/1562 ━━━━━━━━━━━━━━━━━━━━ 284s 182ms/step - accuracy: 0.5008 - loss: 1.6893

Generated Text:

ROMEO: ixbles! We miy
VUblect whow is son well;--
Wour that I shalm
Maser to sherve toed botulius a gased
Whie my be crucur mangipomaen.

BRUTUS:
Is the had corturire.

SICINIUS:
Ly undingly, that them, meings stough' what
A seem it fut desein! The wish of oman face,
Of a ye.
Your shear, come us.
Go minst
```